

The Number That Nobody Explained

How the Icosahedron Solves the Oldest
Riddle of Lepton Physics

*Fundamental Fractal Geometric
Field Theory (FFGFT)*

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<https://github.com/jpascher/T0-Time-Mass-Duality>

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Preliminary Remark

This book tells of a theory that dares to build everything from a single number. Not approximately. Exactly.

It begins with the big picture: what FFGFT is, on which geometry it is built, and why the masses of particles are not free parameters but consequences of a geometry. The first part lays the foundation.

The second part explains how the space and time we observe emerge from the closed four-dimensional torus. This is not a side issue, but the key to the third part.

The third part comes to the icosahedron story: an old puzzle, an unexpected geometric object, and a calculation that closes a forty-year-old open question.

The transitions between the three parts are deliberately kept explicit. Anyone who loses the thread can re-enter at the beginning of the next chapter.

All derivations and verification scripts are in the FFGFT corpus on GitHub (Docs. 205, 270, 285, 293, 338, 352, 364, 367, 368, 369, 370).

Part I

The Foundation: One Theory, One Parameter

Chapter 1

The Big Picture

1.1 An Unusual Starting Point

Most physical theories begin with an equation. Newton begins with $F = ma$. Maxwell with his four equations. Einstein with the equivalence of mass and energy. These are statements about forces and fields that are measured and verified.

FFGFT begins with a **form**. Not with an equation, but with a geometric shape: the four-dimensional torus. This is unusual, and it is worth briefly asking why a form can be a theory.

The answer lies in the fact that physical laws are often **symmetries**. The conservation of energy and momentum follows from time-translation and space-translation invariance — from the fact that the form of natural laws is the same if one moves the clock forward or shifts by one meter. This is a geometric principle, not a mechanical one.

If one now asks which geometry of space can generate all known symmetries of particle physics, one arrives at the torus. Not as an assumption, but as the result of a requirement: which space has exactly the right symmetry groups?

1.2 What ξ Means

The parameter $\xi = 4/30000$ is a ratio: it describes how strong the fractal correction is relative to the base term. Four parts correction to thirty thousand parts base structure. This is tiny — and yet it determines all masses.

The reason lies in **exponentiation**: the mass of a vibrational mode scales as ξ^p with an exponent p that depends on the winding numbers of the mode. Even small differences in p produce large differences in mass. The electron and the tau lepton differ in their exponent by $5/6$ — a difference that is inflated by $\xi^{5/6}$ to a mass ratio of $1 : 3477$.

This is the power of the power law: a small lever, a large effect.

The Fundamental Constant

$$\xi_0 = \frac{4}{30000} \approx 1,33 \times 10^{-4}.$$

It has no unit — just as π has no unit.

FFGFT — Fundamental Fractal Geometric Field Theory — is the attempt to derive all known elementary particles and their interactions from a single geometric premise: that three-dimensional space carries the topological structure of a four-dimensional, fractally structured torus.

The theory thus does not claim to replace the Standard Model. It claims to give it a geometric foundation: the masses that appear in the Standard Model as 25 free parameters are, in FFGFT, not free numbers but eigenvalues of a fixed geometry.

Chapter 2

Why a Torus, and Why Fractal?

2.1 The First Question

The first question that every theory of particle physics must answer is: on what exactly is what vibrating?

The Standard Model has its answer: quantum fields vibrate on flat Minkowski space. This works well computationally, but it is not an explanation — it is a premise.

FFGFT makes a different choice. The reason for this choice is not aesthetic but functional: a torus generates **quantized vibrational modes**, and quantized modes yield discrete masses. The particle spectrum is therefore not an additional input but a consequence of the topology.

2.2 Why Four Dimensions?

An ordinary torus is two-dimensional: the surface of a donut. But physics takes place in four dimensions — three spatial and one temporal. So the torus must be four-dimensional: T^4 .

This means that four independent directions are periodic. Each direction has a radius; the possibility of winding around it generates a quantum number. Four directions, four sets of quantum numbers — this is the origin of the winding numbers.

2.3 Why Fractal?

A pure torus would have perfect periodicity. Nature does not: coupling constants run with energy, masses have a spectrum, there are corrections at all scales.

All of this fits a **fractal** geometry. Not in the sense of Mandelbrot patterns, but in the precise sense of **scale self-similarity**: the same basic structure at different length scales, with a fixed ratio ξ between successive scales.

Chapter 3

What Vibrates: The Modes

3.1 The Image of the String

Imagine a guitar string. It vibrates, and only certain frequencies are allowed: the fundamental frequency and its integer multiples. This is due to the boundary condition: the string is fixed at both ends, and the vibration must be zero at the ends.

On a torus there are no ends. Instead, a different condition applies: the vibration must return exactly to its starting point after one winding. This is the **periodic boundary condition**, and it produces the same as the fixed string: a discrete spectrum of allowed frequencies.

3.2 Not All Modes Are Equal

The D_4 lattice of the torus sorts the modes. Not every combination of winding numbers is equally allowed: the lattice restricts the possible combinations and groups them.

From this grouping arises **trality**: a threefold symmetry that divides the modes into three classes. The first class contains the symmetric modes — the tau leptons and the heavy bosons. The second and third contain the phase-bearing modes — electrons, muons, quarks.

Three classes: three generations. This is not a postulate but a consequence of the lattice geometry.

3.3 Three Families

The triality of the D_4 lattice automatically generates a three-part structure of the winding modes. Not as a postulate, but as a consequence of the lattice geometry: three modes belong together, and there are three such groups. This is the geometric origin of the three lepton generations — and of the three quark generations.

Chapter 4

Where Mass Comes From

4.1 Mass Is Not a Property

In everyday understanding, an object has a mass just as it has a color or a shape — as an inherent property. In physics, this is more complicated.

In the Standard Model there is a mechanism for mass: the Higgs field. The strength of the interaction — the Yukawa coupling — is a free parameter for each particle. 25 particles, 25 parameters. That is not a mechanism, that is a dictionary.

FFGFT has no Higgs mechanism. Mass arises differently.

4.2 Mass as Binding Energy

In FFGFT a particle vibrates as a mode on the torus. The mode is not free — it interacts with the fractal geometry lattice. This interaction has a strength that depends on the winding number of the mode: the more winding numbers, the stronger the binding, the greater the mass.

This yields a formula:

$$m_i = r_i \cdot \xi^{p_i} \cdot \nu,$$

where r_i and p_i come from the winding numbers and $\nu = 246$ GeV is the electroweak scale.

4.3 The Recursion

The fractal geometry has a further consequence: a cascade of corrections following the pattern ξ^n . Each scale corrects

the next by the factor $\xi \approx 1,3 \times 10^{-4}$. This is tiny, but it is not zero.

The recursion explains why the masses of the three generations are so different: electron, muon and tau differ in their winding numbers, and the power law ξ^p produces a strong spread even for small differences in the exponent.

Chapter 5

The Bridge Formula: How Everything Fits Together

5.1 The Relation Between Geometry and Measurement

Physics is the art of translating geometric facts into measured values. The geometry of the torus is abstract; masses in electron volts are concrete. Between the two, one needs a bridge.

In FFGFT this bridge is ξ . On one side stand the winding numbers: integers, from geometry, without unit. On the other side stand masses: in electron volts, measurable, with unit. ξ translates, and $\nu = 246$ GeV sets the scale.

5.2 Why Only One Parameter?

One parameter for all masses is a strong claim. It says: the masses of particles are not independent of one another. They all follow from the same geometric fundamental constant.

This is falsifiable: if one finds a single particle that does not fit the formula $r \cdot \xi^p \cdot \nu$ — and that with integer r and rational p from the lattice structure — the theory is refuted.

5.3 One Parameter for Everything

The astonishing thing about FFGFT is not that it has many parameters. The astonishing thing is that it has a single one.

$\xi = 4/30000$ is dimensionless. From it follow — via the lattice structure of the T^4 torus — all other dimensionless ratios of physics: the mass ratios of leptons, the fine-structure constant, the ratios of coupling constants.

Chapter 6

Finite Fields: The Number Theory Behind the Masses

6.1 What a Finite Field Is

At school one calculates with rational numbers, real numbers, complex numbers. These are infinite fields: between 0 and 1 lie infinitely many numbers.

There are also **finite fields**. The smallest is $\text{GF}(2)$: only two elements, 0 and 1. Addition: $1 + 1 = 0$ (no carries). Multiplication normal. This is the basis of all computer programs: bits.

$\text{GF}(p)$ for a prime p has exactly p elements. $\text{GF}(p^n)$ has p^n elements. $\text{GF}(9) = \text{GF}(3^2)$ has 9 elements; it contains ϕ as an algebraic number. $\text{GF}(81) = \text{GF}(3^4)$ has 81 elements; it contains the fifth root of unity.

6.2 Why Characteristic 3?

In a field of characteristic p , $1 + \dots + 1 = 0$. In $\text{GF}(3)$:

$$1 + 1 + 1 = 0.$$

This has a strange consequence: $x^3 - 1 = (x - 1)^3$ in characteristic 3. So $x^3 = 1$ has only the triple root $x = 1$ — no primitive third roots of unity.

This means: the three itself cannot be a phase in a $\text{GF}(3^n)$ field. The three is neutralized — it must appear as a lattice operation, not as a phase rotation.

6.3 Where the Five Finds Its Place

The fifth root of unity ζ_5 lives in $\text{GF}(81)$. This is because $5 \mid 80 = |\text{GF}(81)^*|$: the multiplicative group of $\text{GF}(81)$ has order 80, and 5 divides 80.

In $\text{GF}(81)$ one can therefore express a rotation by 72° as a phase. This is the channel through which the icosahedron enters the algebraic structure.

Chapter 7

1/α: The Fine-Structure Constant

7.1 The Most Mysterious Number in Physics

Richard Feynman called it “one of the greatest damn mysteries of physics”: the fine-structure constant α . Its reciprocal is $1/\alpha \approx 137,036$ — a dimensionless number that determines the strength of the electromagnetic interaction.

FFGFT derives it. Not exactly — but at a level that is extraordinarily precise for a derivation.

7.2 The Derivation

In FFGFT $1/\alpha$ comes from the Galois structure of the $\text{GF}(3^n)$ fields that algebraically describe the lattice of the torus.

The Fine-Structure Constant

$$\frac{1}{\alpha} = \frac{3700}{27} \cdot K_{\text{frak}}$$

with the fractal correction factor $K_{\text{frak}} = 74/75$.

The number 3700 is the product $4 \cdot 25 \cdot 37 = 2^2 \cdot 5^2 \cdot 37$. The prime factors 2, 5 and 37 are the new primes that appear in the field $\text{GF}(3^{18})$ — that field in which the 37-phase ζ_{37} lives. And 27 is 3^3 , the order of $\text{GF}(27)$.

The result: $1/\alpha \approx 137,036$. Measured value: 137,036. The agreement is better than 10^{-4} .

Chapter 8

What the Theory Does Not Claim

8.1 The Three Most Common Questions

First question: If FFGFT derives everything from one parameter — has it then explained everything? No. A single parameter is not a lack of ambition — it is a strong claim.

Second question: Does FFGFT contradict quantum mechanics? No. FFGFT makes statements about the spectrum of states — what exists. Quantum mechanics makes statements about the dynamics of states — what happens to them. Both are compatible because the levels are separate.

Third question: Are there predictions that can be measured? Yes. The most precise is the Koide formula with $\theta = 2/9$. It is verifiable: if a fourth charged lepton is ever discovered, one can immediately test whether it lies on the Koide curve.

8.2 Three Misunderstandings

This book has a limited focus. It tells the story of the lepton masses and the icosahedron. That is only an excerpt from the overall program.

FFGFT is not a unification in the grand-unified sense. It does not unify electrodynamics, the weak and strong interactions in a single symmetry group. It gives them a common geometric foundation but retains their structural differences.

FFGFT does not replace quantum mechanics. Quantum mechanics describes how one makes measurements on states. FFGFT describes where these states come from. Both levels remain separate.

Part II

The Unrolling: How the Torus Becomes the World

Chapter 9

Four Circles for One World

9.1 An Unexpected Answer

If one asks: How many dimensions does the world have?, the quick answer is: four. Three spatial, one temporal. That has been standard since Einstein.

But that is the observable answer, not the fundamental one. FFGFT asks: How many dimensions does space have before the measurement process turns one of them into time? The answer is likewise four — but that is not the same thing.

9.2 Why Four?

The T^4 torus has four independent circular directions. Why four? The answer is one of the simplest that physics knows: minimal and sufficient.

Minimal: a T^3 would not have enough room for mass and charge as independent conserved quantities. The fourth direction is the mass dimension.

Sufficient: a T^5 or higher would introduce more free parameters than are observable. T^4 is the smallest compact space on which FFGFT manages without redundancy.

9.3 The Central Relation

The key to the mass dimension is a single equation. It says: periodicity times mass equals one.

The Mass Dimension

$$\tilde{T} \cdot m = 1.$$

T is the periodicity of the mass dimension, m the mass of a particle.

From this relation it follows that a massive particle has a time scale — a characteristic periodicity. This is not the same as the lifetime of an unstable particle; it is a topological property of the geometry.

Chapter 10

Two Different Ways of Reducing

10.1 A Seemingly Simple Question

How does one get from four dimensions to one? By omitting three. That sounds trivial. It is not, because “omitting” can mean two very different things.

One can omit a dimension by no longer looking at it: a **true projection**, in which information is lost.

Or one can omit a dimension by translating it into another: a **duality**, in which no information is lost.

This is the decisive difference in the reduction chain of the T^4 torus.

10.2 The Chain Looks Simple — It Is Not

The first step ($T^4 \rightarrow T^3$): The mass dimension is not simply thrown away. It is unrolled into the time direction via the relation $\tilde{T} \cdot m = 1$. This is not a projection; it is a duality. The mode content remains fully preserved — only its visibility changes.

All further steps ($T^3 \rightarrow T^2 \rightarrow T^1 \rightarrow T^0$): These are true projections of spatial directions. Each of them is lossy; information is irretrievably lost. The recursion ξ^n partially compensates for this, but not completely.

10.3 What This Means

Space arises through true projection — through giving up information content. Time arises through duality — through rewriting the same information in a different language. This explains why time has a distinguished direction while the three spatial directions are more symmetric.

Chapter 11

Time as an Unrolled Dimension

11.1 The First Step Is Reversible

The duality decompactification — the unrolling of the mass dimension into time — is information-preserving. This means: if one knows the complete time evolution of a system, one can reconstruct the compact mass dimension from it. The two pictures are equivalent.

This has a measurable consequence: the time evolution of a state must carry a discrete spectral signature — the compact structure survives as a resonance. This resonance encodes itself in the recursion ξ^n : the spacings between energy levels scale with ξ .

11.2 The Holographic Step

The step $T^3 \rightarrow T^2$ — the reduction from three to two spatial dimensions — is special: it has the structure of a holographic projection. The entire three-dimensional information content can be encoded on a two-dimensional surface if the recursion ξ^n compresses the missing dimension.

11.3 A Concrete Image

Imagine the torus as a rolled-up hose: circular in two directions, with two different radii. In four dimensions there are four such circles, with four radii.

The transition $T^4 \rightarrow T^3$ is like cutting a hose open and pulling it straight. The cut circle — the mass dimension — becomes the time axis. The other three circles remain circles: these are the three spatial directions that we observe.

11.4 What Remains

At the end of the chain $T^4 \rightarrow T^3 \rightarrow T^2 \rightarrow T^1 \rightarrow T^0$ stands the observable space: $\mathbb{R}^3 \times \mathbb{R}_t$, three spatial directions and one time direction. This space is not the torus — it is the unrolled image of the torus.

And precisely here — in the unrolled image — the icosahedron acts. Not in the compact torus, where it has no place. But in the space that we see, which arose from the duality and the projections.

This is the bridge to the third part.

Part III

The Icosahedron: The Number That Explains Everything

Chapter 12

The Golden Ratio — An Old Acquaintance

12.1 The Most Beautiful Number in Mathematics

The golden ratio $\phi = (1 + \sqrt{5})/2 \approx 1,618$ has been known since antiquity. It describes the ratio at which a line segment is divided so that the ratio of the whole to the larger part equals the ratio of the larger to the smaller part.

Concretely: if one divides a rectangle so that a square arises and the remainder is again a similar rectangle — then this rectangle has the aspect ratio ϕ . One can repeat this arbitrarily often. This is self-similarity: the fractal property that is contained in the parameter ξ in FFGFT.

12.2 ϕ in the Geometry of the Five

ϕ is closely connected with the five. The cosine of 72 degrees — the interior angle of the regular pentagon — is $(\phi - 1)/2$. The diagonals of a pentagon are in the ratio ϕ to the sides.

12.3 ϕ in the Lepton Masses

That ϕ appears in the lepton masses is therefore not an accident — and also not mere beauty. It is a direct consequence of the icosahedral symmetry that generates the masses.

The redistribution weights p_1 and p_2 , which describe how much of the fivefold rotation goes into the phase-bearing

modes, are in the ratio $p_1/p_2 = \phi^8$. This is exact — no approximation, no fitting.

$\phi^8 \approx 46,98$. The mass ratio $m_\tau/m_e \approx 3477$. The quotient is 74 — a Galois factor from the field $\text{GF}(3^{18})$. That the product $74 \cdot \phi^8$ hits the mass ratio to 0.03 percent is the most precise result of the theory.

Chapter 13

Koide, 1981: Forty Years of Waiting

13.1 A Quiet Discovery

In 1981 the Japanese physicist Yoshio Koide published an observation. No sensation, no headlines — an observation. He had looked at the masses of the three charged leptons and found a relation.

The Koide Formula

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}.$$

Every improvement in the tau mass measurement has confirmed the formula.

13.2 Why the Standard Model Is Silent

The Standard Model has no mechanism that enforces $Q = 2/3$. It has free parameters — the Yukawa couplings — with which one can adjust the masses. But adjusting is not explaining.

In this picture the Koide formula would be an incredible coincidence. Physicists do not like incredible coincidences.

13.3 The Phase and the Question

The formula can be written in another form that shows more structure:

$$\sqrt{m_k} = \sqrt{M} \left(1 + \sqrt{2} \cos \left(\theta + \frac{2\pi k}{3} \right) \right).$$

For $Q = 2/3$ to hold, $\theta = 2/9$ must hold. The question is no longer: Why $Q = 2/3$? The question is: **Why** $\theta = 2/9$?

Chapter 14

The Puzzle

14.1 Three Numbers, One Pattern

If one asks physics students in their third semester how many free parameters the Standard Model has, they usually name three to five. The real number is twenty-five. For every particle mass, every mixing angle, every coupling constant one needs a numerical value that the theory does not explain — that one measures and enters.

Among these twenty-five, three are particularly strange: the masses of electron, muon and tau. Strange because they are not independent of one another. Because a simple formula runs across all three. Because this formula has been known since 1981 — and the Standard Model to this day has no explanation for it.

14.2 A Formula Without Explanation

In the language of the formula: the so-called Koide quotient Q is exactly equal to $2/3$.

That is remarkable in itself. But Koide found more. The three masses obey a single formula that contains only one phase — an angular quantity that determines how the three masses interlock. And this angle is $\theta = 2/9$.

Why two ninths? Koide did not know. The Standard Model of particle physics does not know to this day. The formula works — it hits the masses to twelve decimal places — but no one could say where it comes from.

14.3 What the Number Means

To understand why $2/9$ is so strange, a comparison helps. In physics there are many constants that one can measure but not explain: the fine-structure constant $\alpha \approx 1/137$, the masses of quarks, the mixing angles of neutrinos. The Standard Model accepts these numbers as given and enters them as free parameters — about twenty-five in number. One calls them free because no deeper theory enforces them.

The Koide formula is different. It is not a single number that one could enter somewhere. It is a relation between three masses, and this relation has an internal structure: the phase $2/9$. If one had a deeper theory that explained this phase, it would not replace one free parameter but reveal an entire connection.

Added to this is the precision. Twelve decimal places correspond to a hit to one part in a trillion — better than most experiments in physics. When a formula fits so precisely, it is not an accident.

14.4 What the Standard Model Says

The Standard Model of particle physics is an extraordinarily successful theory. It describes all known elementary particles and their interactions. It has made predictions that have been confirmed to one part in ten billion. But it is silent on the question of why there are exactly three generations of leptons, and it has no mechanism that enforces the Koide formula.

For the Standard Model, the masses of electron, muon and tau are simply what experiment delivers. One measures them, enters them, and moves on. That these three numbers are connected as Koide observed is, for the Standard Model, a coincidence — a very precise but accidental one.

Precisely that is the puzzle: a precision too great for chance, and a formula too simple for the Standard Model, but too strange to be ignored.

Chapter 15

Three Particles

15.1 Three Generations — An Unsolved Question

In nature there are exactly three versions of each charged lepton: electron, muon, tau. Of each neutrino there are three: electron neutrino, muon neutrino, tau neutrino. Of each quark there are three: up/charm/top, down/strange/bottom. Three times three, three times three.

That is not an explanation. That is an observation. The Standard Model enters this threefold number as a fact, without explaining why there are exactly three generations.

FFGFT has an explanation: the **triality** of the D_4 lattice. A symmetry of order three that divides the mode space into three classes. Three classes, three generations. This is geometrically enforced.

15.2 A Puzzle in the Masses

The ratio of the masses of the three charged leptons is $m_e : m_\mu : m_\tau \approx 1 : 207 : 3477$. This is not a uniform distribution.

In FFGFT the reason is the power law: mass scales as ξ^p with the exponent p that depends on the winding number. Even small differences in p produce very large differences in mass at $\xi \approx 10^{-4}$. This explains the hierarchy.

15.3 Three Modes

With triality comes a natural basis: three vibrational modes that exist side by side on the torus. The mathematical name for these are Fourier modes.

The three modes are called v_0, v_1, v_2 . The zeroth mode vibrates symmetrically, the first and second each carry a phase. The tau sits in the symmetric mode, the electron and the muon in the two phase-bearing ones.

With this picture one has explained the three: three generations because the torus has a threefold symmetry. But one does not yet have masses. To get masses, one needs a number — and for this number one needs the icosahedron.

Chapter 16

The Icosahedron

16.1 The Surprising Object

If one asks a physicist which mathematical object explains the lepton masses, one expects as an answer a group like $SU(3)$ or $SO(10)$, perhaps a symmetry principle, a conserved charge. One does not expect an answer from the geometry of ancient Greece.

The icosahedron is one of the five Platonic solids that Plato assigns to the elements in his dialogue *Timaeus*: fire to the tetrahedron, air to the octahedron, water to the icosahedron, earth to the cube, and the cosmos to the dodecahedron. That is not physics, that is mythology.

But the symmetry group of the icosahedron is not mythology. It is a mathematical structure of extraordinary richness — and it is exactly the object that explains the Koide formula.

16.2 A Platonic Solid

The icosahedron is one of the five Platonic solids. It has twenty equilateral triangular faces, thirty edges and twelve vertices. At each vertex five triangles meet. It is the most complicated of the five Platonic solids and the one in which symmetry unfolds most richly.

16.3 The Rotational Symmetries

What makes the icosahedron interesting for physics is not its shape but its symmetry group. The set of all rotations that map the icosahedron onto itself forms a mathematical group: the icosahedral group, called A_5 .

A_5 has sixty elements. That sounds like a lot, but it can be divided nicely: an identity, fifteen rotations by 180° through edge midpoints, twenty rotations by 120° through face centers, and twenty-four rotations by 72° or 144° through vertices.

Among these sixty rotations there are thus elements of order three (the 120° rotations) and elements of order five (the 72° rotations). The icosahedron contains the three and the five under one roof — and in a special way.

16.4 The Smallest Group in Which Three and Five Live Together

$\mathbb{Z}_{15}$ is the cyclic group of order fifteen. It contains elements of order three and of order five. But in $\mathbb{Z}_{15}$ these elements commute with one another: a rotation by three followed by a rotation by five is the same as the reverse.

In A_5 this is different. A 120° rotation followed by a 72° rotation is something different from the reverse order. The three and the five do not commute. This is the technical core — and it is exactly what generates the decisive number.

A_5 is the smallest group in which three and five live together in such a way that they do not commute.

16.5 The Icosahedron and the Torus

Here an important distinction comes in. The torus on which FFGFT is built is a compact, closed object: a four-dimensional torus. On this torus the three acts — as a lattice operation, as triality. But the five has no place on the torus. The symmetry group of the lattice is too small; the five does not share this group.

The icosahedron, by contrast, acts on the space that we observe: the three-dimensional space that arises when one circle of the torus becomes time through the measurement process. That is the “unrolled” picture — the torus unfolded into space and time. On this unrolled space the icosahedron is at home.

The 120° rotation about the axis $(1, 1, 1)$ is exactly the same as the triality of the torus. Without reinterpretation, without renaming: the same mathematical operation, described once as a lattice automorphism of the torus, once as a rotation of the icosahedron in observable space. The three is the same in both pictures.

The five is only in the unrolled picture.

Chapter 17

Three and Five — The Non-Commuting

17.1 A Toy Example

Take a book. Lay it flat on the table, spine up. Now rotate it: first 90 degrees about the vertical axis, then 90 degrees about the horizontal transverse axis. Where is the book now?

Perform the same rotations in reverse order: first the transverse axis, then the vertical axis. The book lands differently.

This is non-commutativity in everyday life. Two rotations, performed in different orders, give different results.

17.2 What Non-Commuting Means

In mathematics an operation that commutes with another is called **commutative**. The addition of integers is commutative: $3 + 5 = 5 + 3$. The multiplication of matrices is generally not.

For the icosahedron this means: the 120° rotation about the $(1, 1, 1)$ axis and the 72° rotation about the $(0, 1, \phi)$ axis — with $\phi = (1 + \sqrt{5})/2$ the golden ratio — give different results in different orders.

17.3 The Mathematics of Non-Commuting

In mathematics a group is called “non-abelian” if it contains non-commutative elements. The group A_5 of icosahedral rotations is non-abelian.

Non-abelian groups are important in physics. The gauge symmetries of the Standard Model — $SU(3)$, $SU(2)$, $U(1)$ — are likewise non-abelian (except $U(1)$). The non-commuting in these groups is the reason for the self-interaction of gluons, which makes the strong nuclear force so strong.

In FFGFT the non-commuting of C_3 (the three) and R_5 (the five) is the reason that a rotation generates a mixing. If both commuted, the electron mode would remain with itself: no mixing, no $2/9$.

17.4 Why This Generates a Number

Imagine the three modes v_0, v_1, v_2 are like three pots. The triality rotation — the three — sorts the pots in a certain way. If one applies the fivefold rotation of the icosahedron to the electron mode v_1 , then the result does not land cleanly back in v_1 . It distributes over all three modes because the rotation does not commute with the sorting.

The shares that fall to the three modes are the redistribution weights. They are fixed — determined by the geometry of the icosahedron — and can be calculated.

The weight that falls to the symmetric mode v_0 is

$$p_0 = \frac{2}{9} = \frac{2}{3^2}.$$

Two ninths. The number that has occupied Koide since 1981.

17.5 Why Exactly 2/9?

The number has a simple explanation if one knows how to count. In the numerator stands the number of non-trivial modes: two (for v_0 is the symmetric, the “trivial” one). In the denominator stands the order of the group squared: $3^2 = 9$. The quotient is $2/9$.

This is not an adjustment, not a fit, not a free parameter. It is a consequence of the group structure: two non-trivial modes, group of order three, absolute square of the projection operator.

17.6 The Whole Table

The fivefold rotation does not only distribute the electron mode. It distributes all three modes. Each row of the following table shows where a mode goes: the first value says how much lands in the symmetric mode v_0 , the second in v_1 , the third in v_2 .

	v_0	v_1	v_2
v_0	5/9	2/9	2/9
v_1	2/9	$(2 + 3\phi)/9$	$(5 - 3\phi)/9$
v_2	2/9	$(5 - 3\phi)/9$	$(2 + 3\phi)/9$

The first row says: of the symmetric mode $5/9$ is preserved, and $2/9$ goes to each of the other two. The amplitudes are $\sqrt{5/3}$, $\sqrt{2/3}$, $\sqrt{2/3}$ — and $5 + 2 + 2 = 9 = 3^2$.

In this one row the five of the icosahedron and the three of FFGFT stand side by side. The non-commuting has produced a number.

In the second and third rows ϕ appears — the golden ratio. It appears where the fivefold rotation connects the two phase-bearing modes, because these modes “see” the five.

The symmetric mode does not see it — and therefore yields a rational number, $2/9$.

Chapter 18

The One Calculation

18.1 A Single Matrix Element

In quantum mechanics one calculates transition probabilities as absolute squares of so-called matrix elements. The question is: if a system is in state A and one applies an operation, how much lands in state B? The answer is a complex number; its absolute square gives the probability.

In FFGFT the relevant question is: if the electron mode v_1 is present and one applies the fivefold rotation R_5 , how much lands in the symmetric mode v_0 ? Physicists write this overlap with a compact notation:

$$A = \langle v_0 | R_5 | v_1 \rangle.$$

Read: how much of R_5 applied to v_1 lands in v_0 . The result is a complex number.

This one matrix element carries the entire Koide formula.

18.2 Modulus and Absolute Square

A is a complex number. Its modulus is $|A| = \sqrt{2}/3$. Its absolute square is $|A|^2 = 2/9$.

Both are needed in the Koide formula — and precisely as follows:

The Koide Formula from the Matrix Element

$$a_k = 1 + 3|A| \cos\left(|A|^2 + \frac{2\pi k}{3}\right).$$

The coefficient before the cosine is $3|A| = \sqrt{2}$. The argument of the cosine contains $|A|^2 = 2/9$ as phase.

Modulus supplies the amplitude, absolute square supplies the angle. The same matrix element in two roles. No free parameter — the matrix element already contains everything.

18.3 What the Koide Formula Says

The formula gives three amplitudes a_0, a_1, a_2 for $k = 0, 1, 2$. The masses of the three leptons are proportional to a_k^2 :

$$m_k = M \cdot a_k^2.$$

The scale M — how large the masses are overall — does not come from the icosahedron. It is the only anchor that the theory needs from outside, and it comes from the other part of FFGFT, the torus ladder.

Everything else — the ratios of the masses, the Koide quotient $2/3$, the phase $2/9$ — follows from the matrix element. To twelve decimal places of precision.

18.4 A New Answer to the Old Question

Koide measured the formula. FFGFT derives it. The question “Why $2/9$?” now has an answer: because $2/9$ is the absolute square of the matrix element that describes how much the fivefold rotation of the icosahedron throws from the electron mode into the symmetric mode.

The answer is so simple that it sounds surprising. But simple does not mean naive in physics — it means that the underlying structure is clean.

Chapter 19

Two Worlds

19.1 The Torus: Compact and Closed

FFGFT is built on a four-dimensional torus. A torus is a closed, finite surface without boundary — like the surface of a donut, but in four dimensions. On this torus fields can wind, and the winding numbers — the wrapping numbers — determine the properties of the particles.

The torus carries a lattice: the D_4 lattice structure. This lattice is the foundation of all integer structures in FFGFT. The triality — the threefold symmetry that explains the three lepton generations — is a property of this lattice.

Crucially: the lattice knows no five. The symmetry group of the D_4 lattice has 1152 elements, and 5 does not divide 1152. The icosahedron — with its fivefold rotations — does not fit geometrically on the torus. This is not a problem but a finding.

19.2 Observable Space: Unrolled

What we observe is not the torus directly. The torus becomes space and time: one circle of the torus becomes, through the measurement process, time. What remains is a three-dimensional space. That is the “unrolled” picture.

In the unrolled picture the icosahedron acts. Here R_5 rotates by 72° , here sits the axis $(0, 1, \phi)$, here the non-commuting generates the number $2/9$.

The three is the same in both pictures. The cyclic permutation $(x, y, z) \mapsto (y, z, x)$ in unrolled space is exactly the same

operation as the triality of the lattice — two names for the same object. Thus the two pictures close together.

The five is only in the unrolled picture. It appears not as a lattice symmetry but as a phase — as an eigenvalue of the rotation that lives in an algebraic field, not in the lattice.

19.3 Why the Five Must Enter as a Phase

This is not one possible reading among others, but enforced. In the mathematics of finite fields that FFGFT uses for the algebraic structure, the following holds: in a field of characteristic 3 there is no primitive third root of unity. The equation $x^3 - 1 = 0$ has modulo 3 only the triple root at $x = 1$. The three can therefore not appear as a phase — it must come as a lattice property.

The five, by contrast, cannot be lattice — it does not divide the lattice group order. It must appear as a phase.

The prime 3 automatically distributes the two symmetries onto two different channels. This is not a choice but a necessity.

19.4 Where They Meet

The two channels — icosahedral phase and lattice arithmetic — touch at a single point: the golden ratio ϕ .

ϕ is at home in both worlds. In the algebraic field $\text{GF}(9)$ — the smallest finite field that contains $\sqrt{5}$ — ϕ lies as a primitive element. This field $\text{GF}(9)$ is a subfield of both the icosahedral field $\text{GF}(81)$ and the arithmetic field $\text{GF}(3^{18})$.

ϕ is the interface. The two worlds meet in ϕ and nowhere else.

Chapter 20

What the Theory Says — and What It Does Not

20.1 Two Classes of Precision

FFGFT makes two kinds of statements about the lepton masses, and it is important to keep the two apart.

The first class are **ratios**: m_μ/m_e and m_τ/m_e . These follow from the geometry of the icosahedron — parameter-free, without free adjustments. The precision is 10^{-3} percent, i.e., one millionth. This lies within the experimental measurement uncertainty of the tau lepton.

The second class are **absolute masses**: how many milli-electronvolts the electron weighs. For this the theory needs an anchor — an external scale that says how large the unit is. This anchor comes from the other part of FFGFT: the torus ladder, which supplies the absolute mass via winding numbers. The precision of this route is about half a percent.

20.2 The Koide Scale and the Torus Ladder

The Koide formula supplies ratios, but also a scale M : the geometric mean of the masses divided by six. This scale is not an independent parameter. It is nothing other than what the torus ladder supplies when one adds up all three masses and divides by six.

This is an important insight: the Koide scale is not a new quantity. It follows from the TO ladder. The icosahedron and the torus ladder do not say two different things — they say

the same thing, once in the language of geometry (more precise) and once in the language of number theory (absolute).

20.3 Where Does the Scale Come From?

The Koide formula supplies ratios. The torus ladder supplies absolute masses — with an external scale $\nu = 246$ GeV. Where does this scale come from?

$\nu = 246$ GeV is the electroweak scale: the vacuum expectation value of the Higgs field. In the Standard Model this is a measured value that is entered into the theory. In FFGFT it is the only external anchor.

Can FFGFT explain ν itself? Yes, and the derivation is in the corpus (Doc. 006, R104). From the FFGFT lattice structure follows $\nu = 248,3$ GeV without fractal correction, or $\nu = 245,6$ GeV with the correction factor $K_{\text{frak}} = 74/75$. The comparison value 246,22 GeV is, strictly speaking, not a direct measured quantity but an SM definition: $\nu \equiv (\sqrt{2} G_F)^{-1/2}$, i.e., the Fermi constant in different clothing (Doc. 351). The agreement is 0,3 % — the same precision class as the other absolute masses of the T0 ladder.

20.4 Theory Plus Convention

If one knows a ratio, one knows the form of the spectrum. If one wants an absolute number, one needs a unit. The unit is a human convention — kilogram, electron volt, solar mass.

FFGFT predicts the form of the lepton mass spectrum to one millionth. That is theory. The absolute size to half a percent is theory plus convention.

Chapter 21

Five Routes to the Same Numbers

21.1 Not One Argument, but Five

A physical theory is more convincing when the same result follows by different routes. The Standard Model has only one route for the lepton masses: measure and enter. FFGFT has five.

21.2 Route 1: The TO Ladder

From the torus winding numbers follows the mass formula $m_i = r_i \cdot \xi^{p_i} \cdot v$. It hits the mass ratios to 0.5 to 1.6 percent — and as the only route also supplies absolute masses.

21.3 Route 2: The Koide Formula with $\theta = 2/9$

From the matrix element $A = \langle v_0 | R_5 | v_1 \rangle$ of the icosahedral group follows the Koide formula, complete and parameter-free. Precision: 10^{-3} percent. This corresponds to the measurement uncertainty of the tau lepton.

21.4 Route 3: Galois Identities

From the orders of $\text{GF}(3^n)$ follow two constraints:

$$\left(\frac{m_\mu}{m_e}\right)^2 = 43200 = 2^6 \cdot 3^3 \cdot 5^2, \quad m_e m_\mu = 54 \text{ MeV}^2.$$

Both are numerically confirmed.

21.5 Route 4: The ϕ Skeleton

From the p values from A_5 follow pure ϕ powers:

$$\frac{p_1}{p_2} = \phi^8, \quad \frac{p_0}{p_2} = 2\phi^4.$$

With ξ correction, m_τ/m_e reaches the precision of the Koide route.

21.6 Route 5: The Connection

Route 5 would derive the scale M (Route 2) from ξ and ν (Route 1). This calculation is still open — see Chapter 23.

21.7 What Each Route Achieves

Route	Input	Output	Precision
1	ξ, ν, r_i, p_i	absolute	0.5–1.6%
2	A	ratios	$10^{-3}\%$
3	$\text{GF}(3^n)$	constraints	0.5%
4	p_j	ratios	0.03%
4b	p_j, ξ	ratios	0.003%

21.8 Two Classes of Precision

Routes 1, 3 and 4a lie at the percent level (0.5–1.6%). Routes 2 and 4b lie at the 10^{-3} percent level and thus within experimental uncertainty. The difference between the classes is exactly the fractal-recursive correction.

Chapter 22

FFGFT and Quantum Mechanics

22.1 What FFGFT Achieves That QM Does Not

Quantum mechanics is extraordinarily successful. It describes interference, entanglement, spin, tunneling — all experimentally confirmed to many decimal places. But it does not explain why the electron has the mass it has. It does not explain why there are exactly three lepton generations. It does not explain why the masses follow the formula $Q = 2/3$. These questions lie outside its domain.

FFGFT gives answers to these questions — not as a supplement to quantum mechanics, but as a deeper geometric foundation. The geometry of the T^4 torus determines which states exist; quantum mechanics describes what happens to these states upon measurement. These are different levels, but FFGFT goes deeper.

22.2 FFGFT Is Deterministic

This is one of the most important points. FFGFT manages without chance.

In standard quantum mechanics the outcome of a single measurement is in principle random: one can predict only probabilities, not the concrete individual case. This is not a lack of information — it is a fundamental principle of the theory.

FFGFT solves this problem geometrically: what looks like chance in the individual event is the projection of a deterministic geometry onto a lower-dimensional observation space. The geometry of the torus determines everything; what we measure as probability is the statistics of many such projections.

22.3 Explanations That QM Does Not Supply

FFGFT explains things that are entered as given facts in standard quantum mechanics and quantum field theory: Why is $\hbar$ just so large? Why does the electron have exactly $-e$ charge? Why are there three generations? Why does Bell inequality violation hold exactly to the measured extent? In FFGFT these questions follow from the geometry — they are not free parameters but consequences.

This is the actual added value: not an alternative description of the same phenomena, but a level of justification that lies beneath.

Chapter 23

The Open Edge

23.1 A Precisely Posed Question

Good science does not end with an answer that closes all questions. It ends with a precisely posed question that shows how far one has come.

The icosahedron explains $2/9$. But it does not explain why $m_\tau/m_e = 74\phi^8$ and not merely ϕ^8 . The factor 74 comes from the number theory of finite fields — from the order of the field $\text{GF}(3^{18})$. And this field lies far beyond the icosahedron.

$\text{GF}(3^{18})$ contains the golden ratio ϕ — because ϕ lies in the subfield $\text{GF}(9)$, and $\text{GF}(9)$ in the subfield of $\text{GF}(3^{18})$. It also contains the order 26 of the lepton structure. But it does not contain the icosahedral phase ζ_5 — for ζ_5 lives in $\text{GF}(81)$, and $\text{GF}(81)$ is not a subfield of $\text{GF}(3^{18})$. This can be proven: 4 does not divide 18.

The icosahedral geometry and the Galois arithmetic — two approaches to the same masses — are thus at home in incompatible algebraic fields. They touch only in ϕ , the golden ratio, which lives as subfield $\text{GF}(9)$ in both worlds.

23.2 The New Primes of $\text{GF}(3^{18})$

$\text{GF}(3^{18})$ contains several quantities from FFGFT: ϕ (via $\text{GF}(9)$), the lepton order 26 (via $\text{GF}(729)$), and the fine-structure number 37.

The primes 19 and 37 are the two that appear anew only at degree 18. Both divide $3^9 + 1 = 19684$. Why 37 supplies the

Galois factor of the masses and not 19 — that is the sharpest open question of the program.

23.3 What This Means

Either there is a higher structure in which icosahedral geometry and Galois arithmetic come together under one roof. Then ϕ would not be the only point of contact but the visible part of a deeper connection.

Or the two approaches are truly independent. Then it would be remarkable — and in need of explanation — that two such different mathematical worlds describe the same physical quantities so precisely.

Both are an open question. But it is a precise open question.

23.4 Why This Is a Good Place to Stop

The open edge at the end of this book is narrow. It lies in an algebraic field that has a name and an order. The question is: does what sits in $GF(81)$ as icosahedral phase, and what sits in $GF(3^{18})$ as Galois order, have the same root — or are they two different sides of the same structure that happen to touch in ϕ ? That is a question that can be answered. One knows what it consists of.

23.5 The Short Version

The icosahedron is the place where three and five do not get along. FFGFT brings the three. The icosahedron adds the five, so that together they produce a product: the matrix element that yields $2/9$.

Koide measured the number. FFGFT derives it.

The remains — why $74\phi^8$, why degree 18, why the golden ratio is the only link between two algebraic fields — still await their explanation. But the question is posed. And it is precise enough to have an answer.

Part IV

The State of Affairs

Chapter 24

What Makes a Good Theory

24.1 Testability

Karl Popper formulated the criterion for a scientific theory: it must be falsifiable. There must be something that would refute it.

FFGFT is falsifiable. A list:

- A fourth charged lepton that does not lie on the Koide curve.
- A mass ratio of quarks that does not follow from $GF(3^n)$ orders.
- A new measurement of the fine-structure constant that deviates far from $\frac{3700}{27} \cdot \frac{74}{75}$.

None of these has occurred so far. The theory has survived all confrontations with experiment.

24.2 Parsimony

A theory is parsimonious if it explains much and assumes little. The Standard Model explains much but assumes 25 parameters. String theory assumes less but so far explains less.

FFGFT assumes one parameter (ξ), a geometric structure (T^4 with D_4 lattice) and an external scale ($v = 246$ GeV). From these three inputs follow the lepton masses to 10^{-3} percent, the fine-structure constant, the triality of the generations. That is parsimony.

24.3 Connectedness

A deep theory connects things that previously seemed unconnected. Special relativity connects space and time. General relativity connects curvature and gravitation. Quantum field theory connects particles and fields.

FFGFT connects the icosahedron with the lepton masses. That is not an obvious connection — a geometric solid of antiquity and the masses of subatomic particles. That this connection is not forced but follows from the non-commuting of two symmetries makes it more than an analogy.

Chapter 25

What Has Been Shown So Far

25.1 What Is Verified

FFGFT is not a closed theory. It is an ongoing program. But some things are shown:

- The mass ratios of the charged leptons follow parameter-free from the A_5 geometry to 10^{-3} percent.
- The fine-structure constant $1/\alpha = 3700/27$ from the $GF(3^n)$ orders of the lattice structure hits at a similar level.
- The torus ladder supplies absolute masses to 0.5 percent — with a single external scale, $\nu = 246$ GeV.

25.2 The Hierarchy of Precision

Different statements of FFGFT have different precisions, and that is not a deficit — it is structure.

10^{-3} percent: Mass ratios of leptons, from the icosahedral geometry. This is better than many experiments can measure.

0.5 to 1.6 percent: Absolute lepton masses from the TO ladder.

10^{-4} : The fine-structure constant from the Galois fields.

The three levels have a common cause: the more a result relies on transcendental functions (cosine), the more precise; the more on algebraic approximations (power laws), the less. This is a structural finding, not an error.

25.3 What Is Open

The complete quark sector structure is laid out in the geometry, but the derivation of the concrete values is still running.

The Galois factors — why $74\phi^8$ and $30\phi^4$ — come from the number theory of finite fields, but their connection to the A_5 geometry is not yet closed. $GF(81)$ and $GF(3^{18})$ touch only in ϕ . Whether this has a common root or is two independent approaches is the sharpest open edge.

25.4 What the Program Means

A geometric foundation for the Standard Model is not a new idea. Kaluza and Klein tried in 1921 to unify electrodynamics and gravitation through a fifth dimension. Superstring theory builds on ten-dimensional manifolds. The commonality: more geometry, more explanation.

FFGFT takes a different path. Not more dimensions, but a more specific lattice structure in four. Not a theory of everything, but a theory of mass.

That is a narrower goal — and a more testable one.

Chapter 26

What Remains

26.1 The Question Behind the Question

This book has explained one number: $2/9$. A number that stood in the data since 1981 and waited for its explanation.

The explanation is: the icosahedron, the group A_5 , the smallest group in which three and five do not commute. A matrix element between two states of this group carries the coefficient $\sqrt{2}$ and the phase $2/9$. The Koide formula follows.

This is not a metaphor. It is a calculation. All steps are verified; the verification scripts are in the repository.

26.2 What This Says About Physics

Sometimes the explanation lies not in new dynamics but in the right geometry. Not: "There is a new force that makes the masses this way." But: "A geometric object with this symmetry cannot generate any other masses."

That is a stronger statement — and one that is falsifiable. If someone finds a lepton with a mass that does not lie on the Koide curve, the explanation is refuted.

Until then it stands.

26.3 The Open Edge — Once More

$GF(81)$ and $GF(3^{18})$ touch in $GF(9)$, i.e., in ϕ . Nowhere else. That is a mathematical fact, not an open point.

The open edge is: why is ϕ the only point of contact? Is there a higher structure in which icosahedral geometry and Galois arithmetic come together under one roof? Or are they two different sides of the secret that meet only accidentally in ϕ ?

That is the question that carries the program forward.

Chapter 27

A Look Back — and Forward

27.1 What This Book Has Told

Three parts, one idea: mass is geometry.

In the first part we saw how a four-dimensional torus with a fractal lattice structure and a single parameter $\xi = 4/30000$ generates the basic structure of particle masses. No free parameters, no Yukawa couplings, no Higgs mechanism as a black box — only geometry and whole numbers.

In the second part we understood how this torus becomes the observable world. The first step — the unrolling of the mass dimension into time — is a duality, not a projection: information-preserving, reversible. The further steps are true projections. This explains why time and space are fundamentally different.

In the third part we told the detective story: a forty-year-old open question, a surprising geometric object, a calculation that ends in $2/9$. The icosahedron as the place where three and five do not get along — and precisely therefore generate a number.

27.2 What Has Not Yet Been Told

A book about physics that ends with an answer is not yet finished. The answer generates new questions — that is the rhythm of progress.

The open edge: $GF(81)$ and $GF(3^{18})$ touch only in $GF(9)$, i.e., in ϕ . Two approaches to the same masses that meet only

in the golden ratio. Whether a deeper unity lies behind this is the most open problem of the theory.

The quarks: the structure is laid out in the torus, but the concrete mass ratios are not yet fully derived.

The absolute masses: the scale $\nu = 246$ GeV is the only external anchor. Whether it itself follows from the geometry is a research question.

27.3 What This Means

Physics is never finished. What counts is the direction.

FFGFT has shown that the lepton masses are not random numbers. They follow from a geometric principle that one can describe, calculate and falsify. That is more than the Standard Model has achieved for these numbers.

Whether FFGFT is the right theory will be decided by experiment. A fourth charged lepton would test it. A more precise tau mass measurement would pose the Koide phase more sharply. A new structure in the Galois fields could explain the 37/19 difference.

The theory has predictions. That is the most important thing.

Chapter 1

For Readers with Mathematics

1.1 The Fundamental Constant

$$\xi = \frac{4}{30000} = \frac{1}{7500} \approx 1,33 \times 10^{-4}.$$

1.2 The Mass Formula (TO Ladder)

$$m_i = r_i \cdot \xi^{p_i} \cdot v, \quad v = 246 \text{ GeV}$$

with $(r_e, p_e) = (4/3, 3/2)$, $(r_\mu, p_\mu) = (16/5, 1)$, $(r_\tau, p_\tau) = (25/9, 2/3)$ from the torus winding numbers.

1.3 The Koide Formula

$$\sqrt{m_k} = \sqrt{M} a_k, \quad a_k = 1 + \sqrt{2} \cos\left(\frac{2}{9} + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2.$$

Koide quotient $Q = (\sum_k m_k) / (\sum_k \sqrt{m_k})^2 = 2/3$ exactly.

1.4 The Derivation from the Matrix Element

$v_k = (1, \omega^k, \omega^{2k})/\sqrt{3}$, $\omega = e^{2\pi i/3}$. R_5 : Rodrigues rotation about $(0, 1, \phi)/\|(0, 1, \phi)\|$, angle 72° .

$A = \langle v_0 | R_5 | v_1 \rangle$: $|A| = \sqrt{2}/3$, $|A|^2 = 2/9$. Thus: coefficient $3|A| = \sqrt{2}$, phase $|A|^2 = 2/9$. Koide formula follows.

1.5 The Redistribution Matrix

$$(|\langle v_j | R_5 | v_k \rangle|^2)_{j,k=0,1,2} = \frac{1}{9} \begin{pmatrix} 5 & 2 & 2 \\ 2 & 2+3\phi & 5-3\phi \\ 2 & 5-3\phi & 2+3\phi \end{pmatrix}.$$

All row and column sums equal 1. Trace $R_5 = \phi$.

1.6 The Two Fields

$\phi \in \text{GF}(9)$ (degree 2, $\phi^4 = -1$, order 8). $\zeta_5 \in \text{GF}(81)$ (degree 4, $5 \mid 80$). $\zeta_{37} \in \text{GF}(3^{18})$ (degree 18, $5 \nmid 18$). $\text{GF}(81) \cap \text{GF}(3^{18}) = \text{GF}(9)$.

1.7 Sources

J. Pascher, FFGFT Corpus, Documents 205, 270, 285, 293, 338, 352, 364, 367, 368, 369, 370.

DOI: <https://doi.org/10.5281/zenodo.22739112>

GitHub: <https://github.com/jpascher/T0-Time-Mass-Duality>

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